

Elastic-Plastic Contact Analysis of a Sphere and a Rigid Flat

L. Kogut

e-mail: mekogut@tx.technion.ac.il

I. Etsion

e-mail: etsion@tx.technion.ac.il

Fellow ASME

Department of Mechanical Engineering,
Technion,
Haifa 32000, Israel

An elastic-plastic finite element model for the frictionless contact of a deformable sphere pressed by a rigid flat is presented. The evolution of the elastic-plastic contact with increasing interference is analyzed revealing three distinct stages that range from fully elastic through elastic-plastic to fully plastic contact interface. The model provides dimensionless expressions for the contact load, contact area, and mean contact pressure, covering a large range of interference values from yielding inception to fully plastic regime of the spherical contact zone. Comparison with previous elastic-plastic models that were based on some arbitrary assumptions is made showing large differences. [DOI: 10.1115/1.1490373]

Introduction

The elastic-plastic contact of a sphere and a flat is a fundamental problem in contact mechanics. It is applicable, for example, in problems such as particle handling ([1]), or sealing, friction, wear, and thermal and electrical conductivity between contacting rough surfaces. Indeed, an impressive number of works on the contact of rough surfaces, that were published so far (see review by Liu et al. [2]), are based on the contact behavior of a single spherical asperity (Bhushan [3]) in a statistical model of multiple asperity contact (Bhushan [4]). Some of these works are restricted to mainly pure elastic deformation of the contacting sphere, e.g., the pioneering work of Greenwood and Williamson [5], which is based on the Hertz solution for a single elastic sphere (e.g., Timoshenko and Goodier [6]). Other works are restricted to pure plastic deformation of the contacting sphere, based on the model of Abbott and Firestone [7], which neglects volume conservation of the plastically deformed sphere.

The works on either pure elastic or pure plastic deformation of the contacting sphere overlook a wide intermediate range of interest where elastic-plastic contact prevails. An attempt to bridge this gap was made by Chang et al. [8] (CEB model). In this model the sphere remains in elastic Hertzian contact until a critical interference is reached, above which volume conservation of the sphere tip is imposed. The contact pressure distribution for the plastically deformed sphere was assumed to be rectangular and equal to the maximum Hertzian pressure at the critical interference. The CEB model suffers from a discontinuity in the contact load as well as in the first derivatives of both the contact load and the contact area at the transition from the elastic to the elastic-plastic regime. These deficiencies triggered several modifications by other researchers. Evseev et al. [9] suggested a uniform pressure distribution, equal to the maximum Hertzian pressure at the critical interference, in the central portion of the contact area, and an elliptical Hertzian distribution outside this portion starting from the maximum pressure and approaching zero at the contact boundary. The authors concluded their paper with a recommendation to find a more general model for the elastic-plastic regime. Chang [10] used an approximate linear interpolation for the elastic-plastic regime by connecting the value of the contact load at yielding inception to that at the beginning of the fully plastic regime. Zhao et al. [11] used mathematical manipulation to smooth the transition of the

contact load and contact area expressions between the elastic and elastic-plastic deformation regimes. Kucharski et al. [12] solved the contact problem of a deformed sphere by the finite element method (FEM) and developed empirical proportional expressions for the contact load and the contact area. Although the authors intended to describe elastic-plastic contact, their results concentrated on the behavior of the sphere deep into the plastic regime. Surprisingly, the mean contact pressure in [12] was, in some cases, higher than the indentation hardness and therefore unreasonable.

The work in [1] employed the finite element method to analyze the contact of two identical spheres, which by symmetry is equivalent to that of one sphere in contact with a frictionless rigid plane. The analysis in [1] was restricted to an aluminum sphere of radius $R = 0.1$ m loaded with a mean contact pressure that never exceeded 2.3 times the material's yield strength.

As can be seen from the literature survey, accurate general solutions for the elastic-plastic contact of a deformable sphere and a rigid flat are still missing. The existing elastic-plastic solutions suffer from several deficiencies caused mainly by assuming some arbitrary contact pressure distribution or an arbitrary evolution of the plastic region inside the sphere. The few existing finite element method solutions are too restricted in terms of materials, geometry, and loading.

It should be noticed here that much research has also been done (mostly by utilizing the finite element method) on the indentation problem of a half-space by a rigid sphere, e.g., [13–16]. However, from the results provided by Mesarovic and Fleck [17] for both a sphere pressed by a rigid flat and a half-space indented by a rigid sphere, deep into the fully plastic regime, it seems that the behavior of these two cases is different. Intuitively, one can see that in the indentation case the radius of the rigid spherical indenter remains constant whereas the curvature of a deformable sphere changes continuously during the deformation. Moreover, the displaced material in the indented half-space is confined by the rigid indenter and the elastic bulk of the half-space. This is quite different from the situation where the displaced material of the deformable sphere is free to expand radially as shown schematically in Fig. 1.

The present research offers an accurate finite element method solution for the elastic-plastic contact of a deformable sphere and a rigid flat by using constitutive laws appropriate to any mode of deformation, be it elastic or plastic. It also offers a general dimensionless solution not restricted to a specific material or geometry.

Theoretical Background

Figure 1 presents a deformable hemisphere, with a radius R , pressed by a rigid flat. The solid and dashed lines show the situ-

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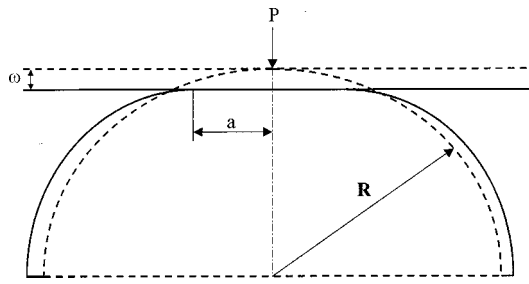


Fig. 1 A deformable sphere pressed by a rigid flat

ation after and before the deformation, respectively. The interference, ω , and contact area with a radius, a , (see Fig. 1) correspond to a contact load, P .

The critical interference, ω_c , that marks the transition from the elastic to the elastic-plastic deformation regime (i.e., yielding inception) is given by (e.g., Chang et al. [8])

$$\omega_c = \left(\frac{\pi K H}{2E} \right)^2 R. \quad (1)$$

The hardness, H , of the sphere is related to its yield strength by $H = 2.8Y$ ([18]). The hardness coefficient, K , is related to the Poisson ratio of the sphere by (Chang et al. [19]) $K = 0.454 + 0.41\nu$. E is the Hertz elastic modulus defined as

$$\frac{1}{E} = \frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2}$$

where E_1 , E_2 and ν_1 , ν_2 are Young's moduli and Poisson's ratios of the two materials, respectively. In the case of the rigid flat $E_2 \rightarrow \infty$.

The Hertz solution for the elastic contact of a sphere and a flat provides the contact load, P_e , and contact area, A_e , for $\omega \leq \omega_c$ in the form

$$P_e = \frac{4}{3} E R^{1/2} \omega^{3/2} = P_c \left(\frac{\omega}{\omega_c} \right)^{3/2} \quad (2)$$

$$A_e = \pi R \omega = A_c \frac{\omega}{\omega_c} \quad (3)$$

where P_c and A_c are the contact load and contact area, respectively, at $\omega = \omega_c$. Note that P_e and A_e can be normalized by P_c and A_c , respectively, to obtain simple exponential functions of the dimensionless interference, ω/ω_c . These functions are independent of the material properties and sphere radius.

Using Eqs. (1)–(3) the mean contact pressure, $p_e = P_e/A_e$, for $\omega \leq \omega_c$ is

$$p_e = \frac{2}{3} K H \left(\frac{\omega}{\omega_c} \right)^{1/2} = p_c \left(\frac{\omega}{\omega_c} \right)^{1/2} \quad (4)$$

where p_c is the mean contact pressure at $\omega = \omega_c$.

For $\omega > \omega_c$ the contact is elastic-plastic and a numerical solution is required to find the relation between ω/ω_c , the contact load, contact area, and mean contact pressure. The finite element method (for example, Refs. [20] and [21]) is commonly used for such a numerical solution where the contact between the sphere and the flat is detected by special contact elements ([22]). A yielding criterion should be adopted in solving elastic-plastic problems. In the present analysis the von Mises criterion, which correlates well with experiments (see Bhushan [3]) was selected as the preferred criterion. A recent example for the finite element method solution to an elastic-plastic contact problem can be found in Liu et al. [23].

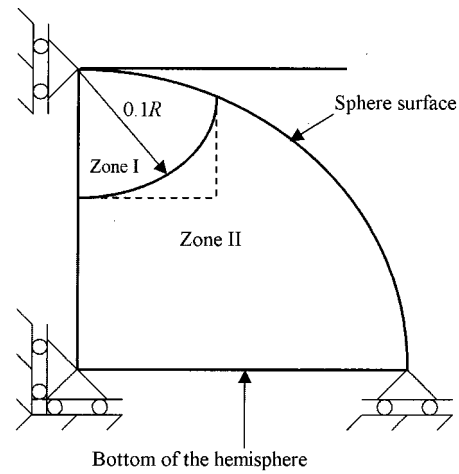


Fig. 2 Model description

Finite Element Model (FEM)

A commercial ANSYS 5.7 package was used to solve the contact problem. The hemisphere, shown in Fig. 2, was modeled by a quarter of a circle, due to its axisymmetry. The rigid flat was modeled by a line. The material of the sphere was assumed elastic-perfectly plastic with identical behavior in tension and compression. Although the model can easily accommodate strain hardening the simpler behavior was selected to allow comparison with existing previous models. A static, small-deformation analysis type was used and justified by comparison with the results of a large-deformation analysis. The von Mises yielding criterion was used to detect local transition from elastic to plastic deformation.

The finite element method numerical solution requires as an input some specific material properties and sphere radius (see [1], for example). However, in order to generalize the present solution and eliminate the need for a specific input, the numerical results were normalized with respect to their corresponding critical values at yielding inception, ω_c , similar to Eqs. (2) and (3). The normalization of the mean contact pressure, p , was done with respect to the yield strength, Y , of the sphere material. The validity of this normalization was tested by solving the problem for several different material properties ($100 < E/Y < 1000$, $\nu = 0.3$) and sphere radii ($0.1 \text{ mm} < R < 10 \text{ mm}$). The dimensionless results of P/P_c , A/A_c , and p/Y versus the dimensionless interference, ω/ω_c , were always the same regardless of the selection of material properties and sphere radius.

The finite element mesh consisted of 225 eight-node quadrilateral axisymmetric elements comprising a total of 714 nodes. High-order elements were selected to better fit the curvature of the sphere. The sphere was divided into two different mesh density zones. Zone I, within a $0.1R$ distance from the sphere tip (see Fig. 2), contained 87% of the nodes and had extremely fine mesh to better handle the high stress gradients in this zone and to achieve good discretization for accurate detection of the contact area radius, a . For this reason the typical mesh size was $0.03a_c$ where $a_c = (R\omega_c)^{1/2}$. Zone II, outside the $0.1R$ distance, had gradual coarser mesh at increasing distance from the sphere tip. The model also contained a single two-dimensional target element laying on the flat and 16 two-dimensional surface-to-surface contact elements on the sphere surface in zone I.

The boundary conditions are presented in Fig. 2. The nodes on the axis of symmetry of the hemisphere cannot move in the radial direction. Likewise the nodes on the bottom of the hemisphere cannot move in the axial direction due to symmetry. Restricting also the radial motion of these nodes did not affect the results of

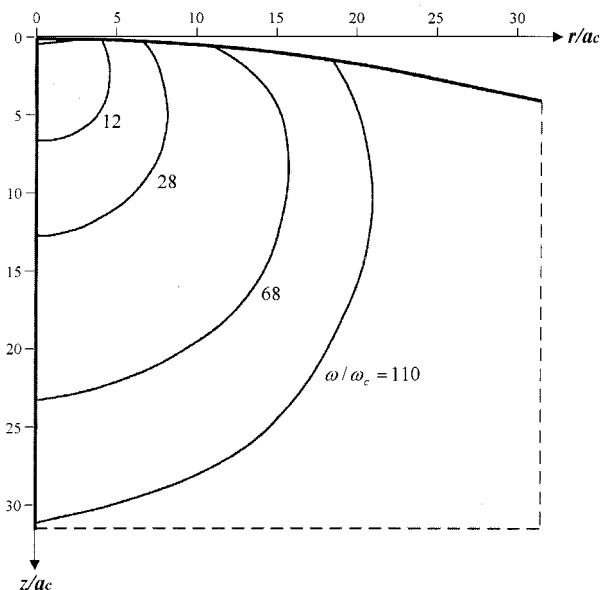


Fig. 3 Evolution of the plastic region in the sphere tip for $12 \leq \omega/\omega_c \leq 110$

the finite element analysis (FEA) since this boundary is very far away from the contact zone and therefore has very little effect on the contact results.

The numerical model was first verified by comparing its output with the analytical results of the Hertz solution in the elastic regime, i.e., for $\omega < \omega_c$. The verification included the contact load, contact area radius, and stress distribution in the contact area and along the axis of symmetry. The difference between the numerical and analytical results was always less than 2.8%. Another verification of the model was done in the elastic-plastic regime (for $1 < \omega/\omega_c < 110$) by increasing the mesh density to 2944 nodes and comparing the results with these obtained with the original 714 nodes. The largest differences in the contact load and contact area were only 1% and 3%, respectively. These two verifications establish the validity of the numerical model with the original mesh to study the behavior of the sphere in the elastic-plastic regime.

Results and Discussion

Figure 3 presents the evolution of the plastic region inside the sphere (within the dashed line frame shown in Fig. 2) for increasing interference values up to $\omega/\omega_c = 110$. The elastic-plastic boundary at each interference is determined by all the nodes with equivalent total strain larger than the yield strain, ε_Y . The axial and radial coordinates in Fig. 3 are normalized by the critical contact radius, a_c . It is interesting to note the larger axial penetration of the plastic region compared to its radial spread. At $\omega/\omega_c = 110$, for example, the plastic region penetrates about $32a_c$ below the contact surface and reaches only about $18a_c$ on the sphere surface.

The evolution of the plastic region at its earlier stages, $\omega/\omega_c \leq 11$, is shown in more details in Fig. 4. Up to $\omega/\omega_c = 6$ the plastic region is completely surrounded by elastic material. At $\omega/\omega_c = 6$ the plastic region first reaches the sphere surface at a radius of about $2.7a_c$. At this point an elastic core remains locked between the plastic region and the sphere surface. As the interference increases above $\omega/\omega_c = 6$ and the plastic region grows, the elastic core gradually shrinks as shown in Fig. 5. The shrinkage rate is very small below $\omega/\omega_c = 30$ and rapidly increases thereafter. The surface of the sphere at the contact region is now divided into three subregions as follows: (I) an inner circular elastic subregion extending radially from the center of the contact until the

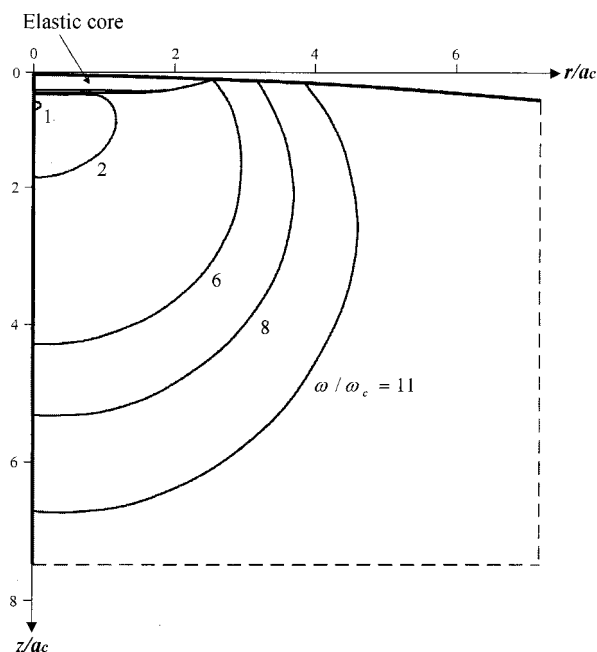


Fig. 4 Evolution of the plastic region in the sphere tip for $1 \leq \omega/\omega_c \leq 11$

edge of the elastic core; (II) an intermediate annular plastic subregion between the edge of the elastic core and the outer front of the plastic region, and (III) an outer elastic subregion thereafter. The evolution of these three subregions on the sphere surface for $\omega/\omega_c \geq 6$ is demonstrated in Fig. 6 that shows the radial locations of the inner and outer elastic-plastic boundaries normalized by the contact area radius, a , as a function of the dimensionless interference ω/ω_c . The horizontal dashed line at $r/a = 1$ indicates the circular boundary of the contact area. From the figure it can be easily seen that below $\omega/\omega_c = 6$ the sphere surface is fully elastic. At $\omega/\omega_c = 6$ the plastic region reaches the sphere surface for the

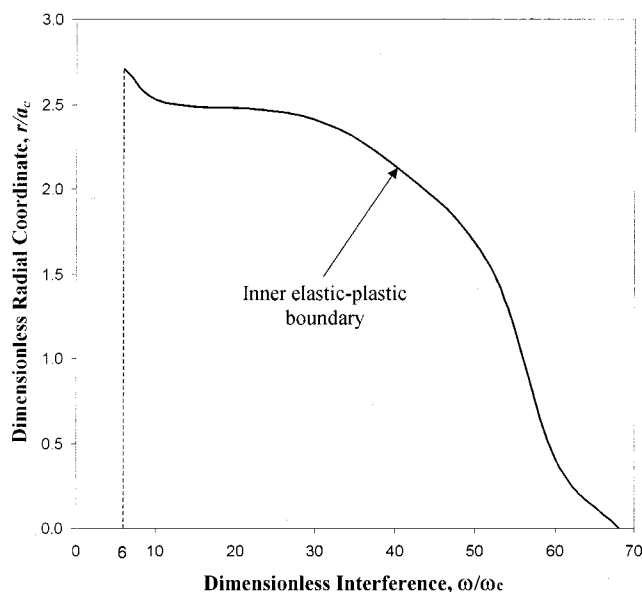


Fig. 5 Dimensionless radial location, r/a_c , of the inner elastic-plastic boundary on the sphere surface showing its shrinkage for $6 \leq \omega/\omega_c \leq 68$

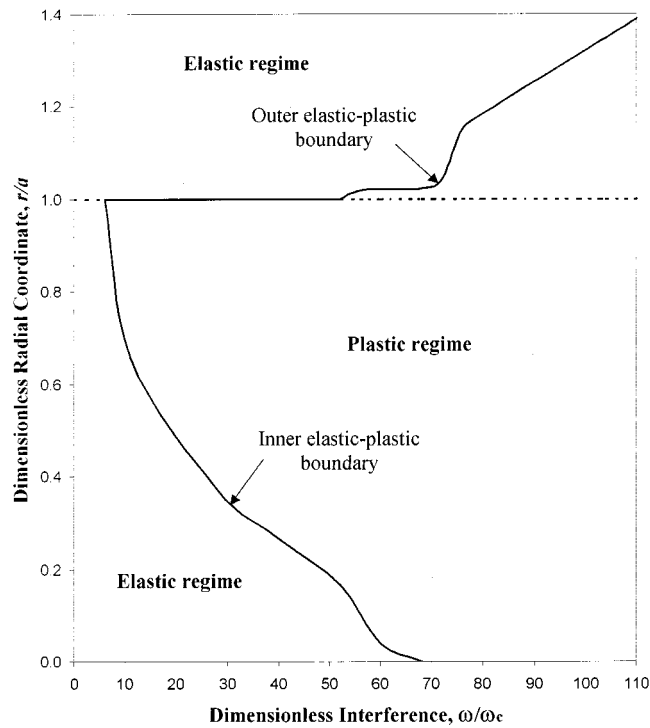


Fig. 6 Radial location of inner and outer elastic-plastic boundaries on the sphere surface for $6 \leq \omega/\omega_c \leq 110$

first time. This occurs very close to the boundary of the contact area, at $r/a = 0.94$. For $6 \leq \omega/\omega_c \leq 56$ the annular plastic subregion remains within the contact area. Its outer boundary, which first reaches the edge of the contact area at $\omega/\omega_c = 6.2$, coincides with that of the contact area while its inner boundary gradually moves towards the contact center as the elastic core shown in Fig. 4 shrinks. For $\omega/\omega_c > 56$ the outer boundary of the annular plastic subregion somewhat exceeds the boundary of the contact area while the inner elastic core continues to shrink and disappears completely at $\omega/\omega_c = 68$. From there on the entire contact zone is plastic and the rate of its radial expansion increases substantially.

From the above discussion it can be seen that the evolution of the elastic-plastic contact can be divided into three distinct stages. The first one for $1 \leq \omega/\omega_c \leq 6$ where the plastic region develops below the sphere surface and the entire contact area is elastic. The second one for $6 \leq \omega/\omega_c \leq 68$ where the contact area is elastic-plastic containing an annular plastic subregion confined by inner and outer elastic ones. The third stage for $\omega/\omega_c > 68$ corresponds to a fully plastic contact area.

Figure 7 presents the results of the mean contact pressure p/Y as a function of the interference, ω/ω_c , that were obtained by the present finite element analysis along with the results from the CEB model ([8]) and from Zhao et al. [11]. When the discrete numerical results of the finite element analysis were curve fitted it became evident that a distinct transition point exists at $\omega/\omega_c = 6$. This is clearly observed in Fig. 7 by the discontinuity in the slope of the finite element analysis results at $\omega/\omega_c = 6$. Apparently, the transition from fully elastic to elastic-plastic contact area, which occurs when the expanding plastic region first reaches the sphere surface, changes the behavior of the mean contact pressure. No similar transition or change was found at $\omega/\omega_c = 68$ that marks the inception of fully plastic contact area when the central elastic core is completely eliminated. The empirical expressions obtained from the curve fitting for the mean contact pressure in the stages that were discussed above are

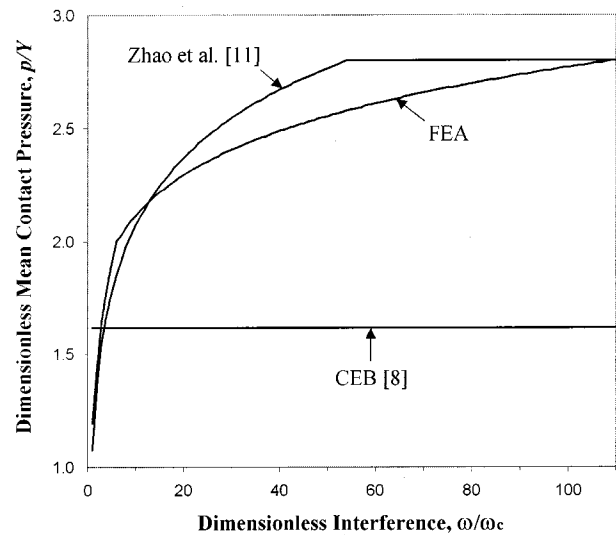


Fig. 7 Dimensionless mean contact pressure, p/Y , as a function of the dimensionless interference, ω/ω_c , in the elastic-plastic regime

$$\left(\frac{p}{Y}\right)_1 = 1.19 \left(\frac{\omega}{\omega_c}\right)^{0.289} \quad \text{for } 1 \leq \omega/\omega_c \leq 6 \quad (5)$$

$$\left(\frac{p}{Y}\right)_2 = 1.61 \left(\frac{\omega}{\omega_c}\right)^{0.117} \quad \text{for } 6 \leq \omega/\omega_c \leq 110. \quad (6)$$

From Fig. 7 it can be seen that the dimensionless mean contact pressure of the finite element analysis at $\omega/\omega_c = 110$ approaches the value $p/Y = 2.8$. This is identical to the ratio between the hardness and yield strength found experimentally for many materials as indicated by Tabor [18]. Hence, the value of p at this point is that of the material hardness, H , and, hence, $\omega/\omega_c = 110$ marks the inception of the fully plastic regime where the mean contact pressure assumes a constant value equals to the material hardness.

The CEB model ([8]) predicts a constant mean contact pressure, which largely underestimates the finite element analysis results except for a small range, $\omega/\omega_c \leq 3$, where it largely overestimates the finite element analysis results. This is one of the limitations of this model as discussed by Evseev et al. [9].

Zhao et al. model [11] predicts p/Y values that are fairly close to the finite element analysis results. The largest deviation of about 9% occurs at $\omega/\omega_c = 54$, which was selected in Ref. [11], based on the work of Johnson [24], as the lowest possible inception of fully plastic regime where $p/Y = 2.8$. Actually the fully plastic regime starts at $\omega/\omega_c = 110$ as can be seen from the finite element analysis results in Fig. 7.

The results obtained by Kucharski et al. [12] cover the range of $175 \leq \omega/\omega_c \leq 2800$ that is very deep into the fully plastic regime and therefore outside the range of interest of the present analysis.

The change in the slope of the mean contact pressure at the transition point $\omega/\omega_c = 6$ is somewhat similar to a typical stress strain curve where a change of slope occurs at the elastic limit. In the spherical contact problem the value $\omega/\omega_c = 6$ is analogous to the critical strain, which corresponds to yielding inception. This point marks the elastic limit of the spherical contact interface. From there on the resistance of the material to increasing strain decreases and eventually disappears at $\omega/\omega_c = 110$.

The finite element analysis results for the dimensionless contact area and contact load are presented in Figs. 8 and 9, respectively, along with the results of Refs. [8] and [11]. The corresponding empirical expressions obtained from curve fitting of the finite element analysis numerical results in the various stages of the evolution of the elastic-plastic contact are

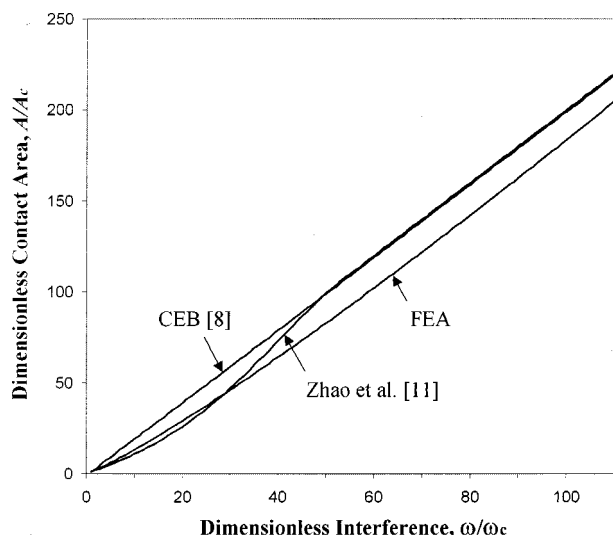


Fig. 8 Dimensionless contact area, A/A_c , as a function of the dimensionless interference, ω/ω_c , in the elastic-plastic regime

$$\left(\frac{P}{P_c}\right)_1 = 1.03 \left(\frac{\omega}{\omega_c}\right)^{1.425}$$

$$\left(\frac{A}{A_c}\right)_1 = 0.93 \left(\frac{\omega}{\omega_c}\right)^{1.136} \quad \text{for } 1 \leq \omega/\omega_c \leq 6 \quad (7)$$

$$\left(\frac{P}{P_c}\right)_2 = 1.40 \left(\frac{\omega}{\omega_c}\right)^{1.263}$$

$$\left(\frac{A}{A_c}\right)_2 = 0.94 \left(\frac{\omega}{\omega_c}\right)^{1.146} \quad \text{for } 6 \leq \omega/\omega_c \leq 110. \quad (8)$$

The accuracy of the curve fitting for Eqs. (7) and (8) was better than 97% throughout the range of ω/ω_c .

From Fig. 8 it is clear that the contact area obtained by the CEB model ([8]) overestimates the finite element analysis results. The largest difference is 56% at $\omega/\omega_c = 4$. This difference diminishes as the interference increases and for $\omega/\omega_c = 110$ it becomes less

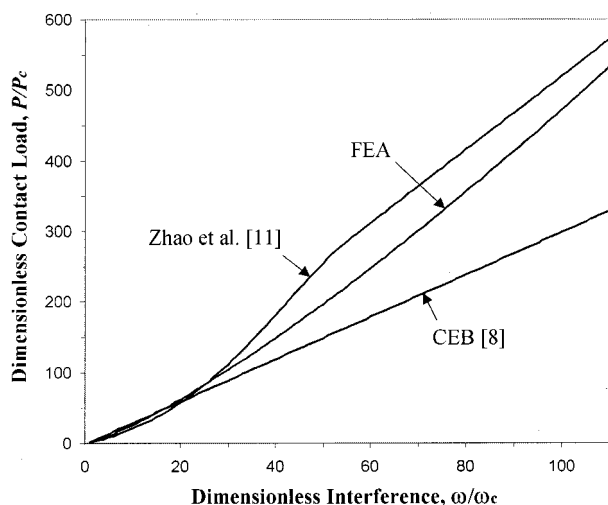


Fig. 9 Dimensionless contact load, P/P_c , as a function of the dimensionless interference, ω/ω_c , in the elastic-plastic regime

than 7%. The reason for the larger deviation at smaller interferences is that the CEB model assumes volume conservation of the entire sphere tip for $\omega/\omega_c \geq 1$. This in fact is equivalent to assuming fully plastic regime of the entire sphere tip as soon as the critical interference is reached. From Fig. 3 it is clear that the plastic region develops gradually with increasing interference and only for very large interferences the entire asperity tip is plastically deformed.

The Zhao et al. [11] results underestimate the finite element analysis ones by up to 18% at $\omega/\omega_c = 10$ and overestimate them by up to 20% at $\omega/\omega_c = 51$. The Zhao et al. model assumes fully plastic sphere tip at $\omega/\omega_c = 54$. From this point on the contact area is calculated from the geometrical intersection of the flat with the original profile of the sphere according to Abbott and Firestone [7]. This is also true for the CEB model, which therefore predicts the same results at large interferences.

At $\omega/\omega_c = 110$ the contact area based on the Abbott and Firestone approximate calculation is only 7% higher than the more accurate result of the finite element analysis. It seems therefore, that the Abbott and Firestone model is a relatively fair approximation for the contact area in the fully plastic regime.

Figure 9 presents the contact load P/P_c versus the interference ω/ω_c . The contact load obtained by the CEB model ([8]) clearly differs from the finite element analysis results. It overestimates the finite element analysis results at small interferences, by up to 62% at $\omega/\omega_c = 2$, and underestimates these results by up to 38% at $\omega/\omega_c = 110$. This is due to a combination of the very inaccurate assumption of constant mean pressure and too large contact area in [8] as shown in Figs. 7 and 8. Contrary to the CEB model, the contact load obtained by Zhao et al. underestimates the finite element analysis results at small interferences (21% at $\omega/\omega_c = 7$) and overestimates these results at large interferences (about 30% at $\omega/\omega_c = 52$).

Since the model is general enough to accommodate material behavior other than elastic-perfectly plastic, various levels of linear isotropic strain hardening were also investigated. In the extreme case of a very large tangent modulus that is $0.1E$, the difference in the results, compared to the present elastic-perfectly plastic case, was less than 20%. In fact for $\omega/\omega_c \leq 20$ the maximum difference was less than 4.5%. For most practical materials the tangent modulus is less than $0.05E$ hence, the difference in the results is much smaller and the present case can be considered a general elastic plastic one.

It is interesting to compare some features of the present contact problem of a deformable sphere and a rigid flat with these of the half-space indented by a rigid sphere. The fully plastic regime in indentation starts at $A/A_c = 113.2$ according to Francis [25], and at $P/P_c \approx 360$ according to Johnson [24]. The corresponding finite element analysis results for fully plastic deformable sphere at $\omega/\omega_c = 110$ are $A/A_c = 205$ and $P/P_c = 534$. Clearly the two problems exhibit different behavior. The indented half-space yields more easily than the pressed sphere. This is probably due to the greater resistance to radial expansion that is imposed on the deflected material in the case of the indented half-space as compared to the case of the deformable sphere.

Conclusion

The elastic-plastic contact problem of a deformable sphere and a rigid flat was solved by the finite element method considering the actual constitutive laws for the relevant regime of deformation. Hence, the present model is much more accurate than previous ones that relied on unrealistic assumptions regarding the contact pressure distribution or evolution of the plastic region above the critical interference. By properly normalizing the contact load, contact area, and mean contact pressure, the present model provides simple analytical expressions that extend the classical Hertz solution up to a fully plastic contact.

It was found that the evolution of the elastic-plastic contact can be divided into three distinct stages. The first one for $1 \leq \omega/\omega_c$

≤ 6 where the plastic region develops below the sphere surface and the entire contact area is elastic. The second one for $6 \leq \omega/\omega_c \leq 68$ where the contact area is elastic-plastic, and the third stage for $\omega/\omega_c > 68$ corresponds to a fully plastic contact area.

The numerical results of the present finite element analysis were normalized in a way that allowed a general solution that is independent of specific material and radius of the sphere. Dimensionless expressions for the mean contact pressure, contact load and contact area were derived for a large range of interference values up to $\omega/\omega_c = 110$.

A change in the behavior of the mean contact pressure was observed at $\omega/\omega_c = 6$, which marks the elastic limit of the contact area. The interference $\omega/\omega_c = 110$ marks the inception of fully plastic regime where the mean contact pressure becomes equal to the material hardness.

A comparison of the present results with the results of previous elastic-plastic models as well as with these of indentation models showed substantial differences.

Acknowledgment

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Nomenclature

- a = radius of contact area
- A = contact area
- E = Hertz elastic modulus
- $E_{1,2}$ = Young's moduli
- H = hardness of the sphere
- K = hardness factor, $0.454 + 0.41\nu$
- P = contact load
- p = mean contact pressure, P/A
- R = radius of the sphere
- Y = yield strength of the sphere
- ν = Poisson's ratio of the sphere
- $\nu_{1,2}$ = Poisson's ratio
- ω = interference

Subscripts

- c = critical values
- e = elastic contact

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